

Course Overview

This beginner-friendly MERN Stack course introduces full-stack web development using MongoDB, Express.js, React.js, and Node.js. By the end of the course, students will be able to build and deploy complete web applications from scratch.

Module 1: Web Development Fundamentals

- Introduction to Web Development
- How the Internet Works
- HTTP & HTTPS Basics
- Client vs Server Architecture
- Introduction to Git & GitHub

Module 2: HTML & CSS Basics

- HTML Structure & Semantic Tags
- Forms and Inputs
- CSS Selectors & Box Model
- Flexbox & Grid Layout
- Responsive Design Principles

Module 3: JavaScript Essentials

- Variables, Data Types & Operators
- Functions & Scope
- Arrays and Objects
- DOM Manipulation
- ES6+ Features (Arrow Functions, Promises, Modules)

Module 4: Backend with Node.js

- Introduction to Node.js
- Node Package Manager (NPM)
- Creating Servers using HTTP Module
- File System Operations
- Environment Variables

Module 5: Express.js Framework

- Routing in Express
- Middleware Concepts
- REST API Development
- Error Handling
- Authentication Basics

Module 6: MongoDB Database

- NoSQL Concepts
- MongoDB Installation & Setup
- CRUD Operations
- MongoDB Atlas
- Mongoose ODM

Module 7: Frontend with React.js

- Introduction to React
- JSX & Components
- Props & State
- Hooks (useState, useEffect)
- React Router
- API Integration

Module 8: Full MERN Integration

- Connecting React with Express APIs
- Authentication (JWT Basics)
- Project Structure
- Environment Configuration

Module 9: Deployment & Version Control

- Git Workflow
- Deploying Backend
- Deploying Frontend
- Environment Variables in Production

Module 10: Final Project

- Build a Full-Stack MERN Application

- User Authentication System
- CRUD Dashboard
- Deployment & Presentation

Learning Outcomes

- Understand full■stack web development concepts
- Build RESTful APIs using Node.js and Express
- Create dynamic UIs using React
- Work with MongoDB databases
- Deploy full MERN applications